

Erik Mauricio

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EDUCATION

Cornell University

Ithaca, NY

Bachelor of Arts in Computer Science and Information Science Minor – GPA: 3.47/4.0

Aug. 2024 – May 2028

- **Relevant Coursework:** Object-Oriented Programming & Data Structures, Discrete Structures, Information Retrieval, Artificial Intelligence, Full-Stack Development, Linear Algebra

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Flask, FastAPI, Tailwind CSS, Socket.io, NumPy

Tools & AI: PostgreSQL, MongoDB, Docker, Git, REST APIs, Claude Code, OpenAI API, RAG, TF-IDF, Word2Vec

EXPERIENCE

Cornell Hack4Impact

Ithaca, NY

Software Developer

Aug. 2025 – Present

- Developed REST API endpoints for restaurants, Community-Based Organizations (CBOs), and meal-provider relationships, joining data across 3 relational database tables to power dynamic sidebar and popup components for 204 partner organizations.
- Integrated Mapbox and Geocoding APIs to convert street addresses to coordinates, rendering color-coded markers for 135 CBOs and 69 restaurants across 5 NYC boroughs serving 30,000 meals/week.
- Designed interactive CBO-Restaurant relationship visualization using Deck.gl, rendering curved arrows to trace meal supply chains and a priority-ordered tag system highlighting underserved communities by poverty threshold.

Cornell University

Ithaca, NY

Undergraduate Teaching Assistant

Aug. 2025 – Present

- Led weekly lab sections and office hours for 20+ students in a full-stack development course (React, Node.js, Express.js, MongoDB), facilitating hands-on exercises and debugging sessions.
- Evaluated assignments and provided individualized feedback on 50+ student submissions, improving code quality and reinforcing core full-stack development concepts.

ThinkNeuro, LLC

Remote

Software Engineering Intern

June 2025 – Aug. 2025

- Developed a B2B license dashboard using Streamlit and Python to track student enrollment, kit shipments, and engagement across partner institutions.
- Automated CSV data ingestion and cleaning pipelines with Pandas, replacing a manual workflow and automatically flagging inactive records and data quality issues.

PROJECTS

Sports Team Finder | *Python, Flask, TF-IDF, Cosine Similarity, RAG, SVD*

Feb. 2025 – Present

- Built a full-stack semantic team search engine over 6.3M Reddit comments and Wikipedia articles, indexing 368 teams across 6 sports with a 416K-term TF-IDF vocabulary.
- Designed a search pipeline combining SVD, cosine similarity, and Word2Vec query expansion, paired with an LLM RAG layer for query rewriting, result re-ranking, and plain-English explanation of results.

Tiempos Perfectos | *React, MongoDB, Express.js, Node.js, Tailwind CSS, Socket.io*

June 2025 – Aug. 2025

- Developed a full-stack Spanish learning platform with 3 practice modules (grammar, reading, real-time conversation) spanning 5 verb tenses and 300+ exercises, with dynamic filtering by tense, difficulty, and question count for personalized AP Spanish exam preparation.
- Integrated real-time speech recognition and AI conversation features using OpenAI GPT-4 and Socket.io, supporting 3 adaptive difficulty levels and simulating AP Spanish speaking conditions.

LEADERSHIP

Society of Professional Hispanic Engineers.

Ithaca, NY

Corporate Chair

May 2026 – Present

- Manage sponsorship outreach and relationships with 20+ companies while coordinating recruiting events, professional development initiatives, and corporate communications for Cornell SHPE members.